

# Can Data Scientists be Replaced by Automation?



The advent of AI, automation and smart bots triggers the question: Is it possible that data scientists will become redundant in the future? Are they indispensable? The ideal approach appears to be automation complementing the work data scientists do. This would better utilise the tremendous data being generated throughout the world every day.

**D**ata scientists are currently very much in demand. But there is the question about whether they can automate themselves out of their jobs. Can artificial intelligence replace data scientists? If so, up to what extent can their tasks be automated? Gartner recently reported that 40 per cent of data science tasks will be automated by 2020. So what kind of skills can be efficiently handled by automation? All this speculation adds fuel to the ongoing 'Man vs Machine' debate.

Data scientists need a strong mathematical mind, quantitative skills, computer programming skills and business acumen to make decisions. They need to gather large unstructured data and transform it into results and insights, which can be understood by laymen or business executives. The whole process is highly customised, depending on the type of application

domain. Some degree of human interaction will always be needed due to the subjective nature of the process, and what percentage of the task is automated depends in the specific use case and is open to debate. To understand how much or what parts can be automated, we need to have a deep understanding of the process.

Data scientists are expensive to hire and there is a shortage of this skill in the industry as it's a relatively new field. Many companies try to look for alternative solutions. Several AI algorithms have now been developed, which can analyse data and provide insights similar to a data scientist. The algorithm has to provide the data output and make accurate predictions, which can be done by using Natural Language Processing (NLP).

NLP can be used to communicate with AI in the same way that laymen interact with data scientists to put

forth their demands. For example, IBM Watson has NLP facilities which interact with business intelligence (BI) tools to perform data science tasks. Microsoft's Cortana also has a powerful BI tool, and users can process Big Data sets by just speaking to it. All these are simple forms of automation which are widely available already. Data engineering tasks such as cleansing, normalisation, skewness removal, transformation, etc, as well as modelling methods like champion model selection, feature selection, algorithm selection, fitness metric selection, etc, are tasks for which automated tools are currently available in the market.

Automation in data science will squeeze some manual labour out of the workflow instead of completely replacing the data scientists. Low-level functions can be efficiently handled by AI systems. There are many